

Multi-agent distributed formation control using local information measurements

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Abstract This paper investigates formation control in multi-agent systems to adjust deformation and scaling for obstacle avoidance using local information. A well-known issue with angle-based formation control is the existence of size ambiguity. Introducing additional controlled variables can potentially mitigate this problem. Therefore, unlike purely angle-based formation control methods that can only adjust deformation, our approach can also control scaling by introducing distance measurements. Specifically, all agents use their local coordinate systems and sense neighboring agents to obtain angle measurements, with only one agent measuring the distance to its neighbor. In the proposed formation control scheme, angle measurements are used to control the desired shape, while distance measurements regulate the formation scaling. Different formation control strategies for triangular and polygonal configurations are discussed, utilizing angle rigidity to ensure the stability of the final shape in polygonal configurations with only one distance measurement used for formation scaling. Additionally, by relaxing the condition of angular rigidity and utilizing the concept of formation rigidity, we propose a formation determination method to construct a rigid square configuration, in which a pair of distances measured by one agent are used to control the formation's scaling. Finally, using a simplified model of a quadrotor unmanned aerial vehicle (QUAV), simulation results validate the effectiveness of the proposed formation control scheme through several examples.

Keywords angle measurements, distance measurements, formation control, rigid formation, multi-agent systems

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1 Introduction

In recent years, collaborative control of multi-agent systems has attracted widespread attention, with formation control issues receiving extensive research due to their significance across multiple application domains [1, 2]. Examples include unmanned aerial vehicle (UAV) search and rescue [3], collaborative transport of ground robots [4], and satellite formation flying [5].

A key focus is addressing which variables need to be sensed and actively controlled to achieve desired formations

in multi-agent systems, as mentioned in ref. [6]. Sensors used for formation control typically include the global positioning system (GPS), radar, and cameras, which can measure positions, inter-agent distances, and bearings, respectively. The existing formation control methods are categorized based on the sensors and control variables used, including position-based, displacement-based (consensus-based) [7, 8], direction-based, distance-based, and angle-based approaches. However, a disadvantage of position-based formation control is that it typically requires a global coordinate system shared among the agents [9]. In displacement-based methods, agents measure the relative positions of their neighbors in a common reference coordinate system, necessitating

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that all agents have a shared sense of direction [10, 11]. This implies that the local coordinate frames of all agents need to be aligned [12]. Achieving perfect alignment of all agents' coordinate systems is challenging due to the presence of sensor measurement noise and interference. When there is a misalignment in the agents' coordinate frames, the formation shape specified by relative positions may be distorted and exhibit non-zero translational velocity [13, 14]. In ref. [15], the theory of bearing rigidity was established, but it still requires agents to know a global reference frame. In contrast, distance/angle-based formation methods that rely solely on local information from multi-agent systems have become popular, even though they increase the complexity of their design and analysis.

Over the past few decades, distance rigidity has been extensively studied both as a mathematical topic in graph theory [16] and as an engineering problem in the application of multi-agent system formations. Distance rigidity [17] is defined based on the property that distances within a multi-point framework remain invariant under translational and rotational motions. Without requiring coordinate frame alignment, the desired formation shape described by distances was achieved using local relative position measurements in ref. [18] by employing distance rigidity theory. Moreover, a theory based on bearing rigidity was proposed [15, 19]. Agents maintain their relative orientation with respect to their neighbors to ensure the stability of the formation. However, methods solely based on bearings become ambiguous due to the lack of measurement for formation size. To address this, a bearing-based approach within the leader-follower framework was introduced [20, 21], but it requires that the maneuvering strategies of two leaders be pre-planned. With the advancement of sensor technology, angle measurements can be easily obtained through omnidirectional radar or vision-based cameras, which are cheaper, more reliable, and more accessible than distance measurements [22, 23]. Since angle measurements are scalars and do not change with motion, including translation, rotation, and scaling [24, 25], the theory of angle rigidity was established in ref. [26]. By assigning appropriate angle constraints to the agents, the desired formation shape can be achieved. However, a resulting issue is that the angles do not change with the scaling of the formation. When relying solely on angle information for control, the size of the formation becomes ambiguous when the multi-agent system stabilizes. To address this issue, a control algorithm for multi-agent systems was designed in ref. [27], which determines the formation shape through angle constraints while still using relative position measurements in the global coordinate system to scale the formation. Recently, the formation control method combining angle and distance has been studied in refs. [28, 29], and a formation control scheme with directed graph distance measurements and angle constraints has been proposed. However, these

methods impose complex directed graph distance constraints and angle measurements, increasing the difficulty of formation rigidity design. In many cases, it is necessary to manipulate the formation scale, such as to avoid obstacles [30], and the variation in pairwise distances between agents during scaling complicates the control algorithm.

Motivated by the work above, we propose a formation scaling control law based on angle and distance to address formation maintenance during obstacle avoidance. This approach requires each agent to measure only the angle information of its neighbors, with only one agent responsible for measuring distance information. Angle rigidity ensures the stability of the final formation shape, while distance control manages the scaling of the formation. The contributions of this paper are summarized as follows.

(1) Compared with ref. [26], this addresses the issue of ambiguity in controlling the formation size using only angle information. In ref. [26], when the formation system is stable, all angle constraints are satisfied. However, because angles do not change with scaling motions, the overall size of the formation is ambiguous and cannot be controlled arbitrarily. Additionally, to relax an angular rigidity constraint in ref. [26], we discuss a method using a square formation case to control the scaling by weighting a pair of distances in two different directions.

(2) Different from the rigidity formation methods that combine angle and distance-directed graph in refs. [28, 29], the scaling transformation of the formation in this paper is controlled by a single agent, eliminating the need to change a large number of desired distances between different agents and thereby simplifying the complexity of the control algorithm during the process of expansion and contraction motion.

(3) This work represents a preliminary step in addressing the problem of formation maintenance, deformation, and scaling control based on local information. Due to the difficulty in constructing a Lyapunov function, we have currently proven the local convergence of the closed-loop nonlinear formation under double integrator dynamics. In view of this, we provide sufficient conditions to ensure the stability of the double integrator formation, which offers valuable guidance for future research.

The remainder of this paper is organized as follows. Section 2 introduces the agent formation control problem. Section 3 presents the main results with different configurations. Section 4 shows the simulation results examining the performance of the proposed formation control law.

Notations: An undirected graph, denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, consists of a vertex set $\mathcal{V} = \{1, 2, \dots, n\}$ and an edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ with $m = |\mathcal{E}|$. The set of neighbors of vertex i is denoted as $\mathcal{N}_i \triangleq \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$. An orientation of an undirected graph is the assignment of a direction to each edge. An oriented graph is an undirected graph together with

an orientation.

2 Problem formulation

For the n mobile agents moving in the plane, we define the set of agent indexes as $\mathcal{V} = \{1, 2, \dots, n\}$. The dynamics of each agent i can be described as

$$\begin{cases} \dot{\mathbf{p}}_i = \mathbf{v}_i, \\ \dot{\mathbf{v}}_i = \mathbf{u}_i, \end{cases} \quad \forall i \in \mathcal{V}, \quad (1)$$

where $\mathbf{p}_i \in \mathbb{R}^2$ denotes the position of agent i in the fixed global coordinate system, $\mathbf{v}_i \in \mathbb{R}^2$ is the velocity and $\mathbf{u}_i \in \mathbb{R}^2$ is the control input.

The following assumptions are made for all agents to achieve the desired formation configuration.

Assumption 1 All agents are already on the same plane when the formation needs to deform and scale.

Assumption 2 Each agent is equipped with a sensor to measure LOS (Line of Sight).

Assumption 3 The initial positions of the agents are independent and non-collinear.

Assumption 1 ensures that the formation is in a two-dimensional plane. Assumption 2 requires each agent to measure the directional angle of its neighbor with respect to its local coordinate system. Assumption 3 guarantees that the angle bisector direction exists and is unique.

Under Assumptions 1 and 2, each agent i can measure the angle $\theta_{ij} \in [0, 2\pi)$ relative to $j \in \mathcal{N}_i$ in its local coordinate frame $\sum_i$, as shown in Figure 1. The local coordinate system satisfies the right-hand frame. Then, the angle α_i in the x_i - y_i plane can be given by

$$\alpha_i = \theta_{ij} - \theta_{ik}, \quad (2)$$

where $\alpha_i \in [0, 2\pi)$ is the interior angle of the triangle formed by j , i and k .

As shown in Figure 1, the agent i can measure the LOS angle of its neighbors. Then, the interior angle α_i can be calculated by

$$\cos \alpha_i = \mathbf{z}_{ij}^\top \mathbf{z}_{ik}, \quad (3)$$

where $\mathbf{z}_{ij} = \frac{\mathbf{p}_j - \mathbf{p}_i}{\|\mathbf{p}_j - \mathbf{p}_i\|}$, which starts from i , points toward j .

Let $\alpha_i^* \in (0, \pi)$ be the desired steady-state angles of the formation as shown in Figure 2(a). The angle-based formation problem becomes the realization of the desired angle constraint:

$$\lim_{t \rightarrow +\infty} (\alpha_i(t) - \alpha_i^*) = 0. \quad (4)$$

Definition 1 (Angle rigidity and global angle rigidity, ref. [26, Definition 2, Definition 3])

We give our description of the uniqueness of the framework based on graph theory.

Definition 2 (Similarity and congruence) If two frameworks $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1, \mathcal{W}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2, \mathcal{W}_2)$ satisfy the following conditions: (1) They have the same topological structure (i.e., there exists a graph isomorphism $f: \mathcal{V}_1 \rightarrow \mathcal{V}_2$, such that for every edge $(i, j) \in \mathcal{E}_1$, there is an edge $(f(i), f(j)) \in \mathcal{E}_2$). (2) For each pair of corresponding edges $e_{1i} \in \mathcal{E}_1$ and $e_{2i} \in \mathcal{E}_2$, their weights satisfy $w_{1i} = k \cdot w_{2i}$, where k is a fixed proportional constant; then it can be said that the two frameworks are similar if $k \neq 1$, denoted as $\mathcal{G}_1 \sim \mathcal{G}_2$ and we say that two frameworks are congruent if $k = 1$, that is, $\mathcal{G}_1 \triangleq \mathcal{G}_2$.

However, a problem arises when relying solely on the constraints of eq. (4); it cannot effectively control the formation scale. For instance, we observed that the angles used to constrain the configuration in Figure 2(a) and (b) both satisfy eq. (4), but they do not determine the distances between the agents. When the system is stable, it satisfies the angular rigidity as defined in Definition 1, but the actual framework $\mathcal{G}$ is not congruent with the desired framework $\mathcal{G}^*$. In other words, achieving the desired shape may result in the agents' formation scale being either too large or too small, that is, similarity as defined in Definition 2.

A triangle can be uniquely determined given its three angles and any one side. Therefore, for triangular formations, we can achieve the expansion and contraction of the

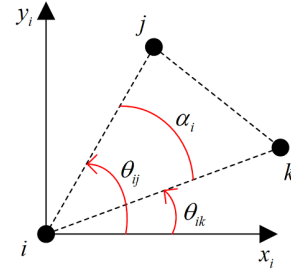


Figure 1 (Color online) The angle measurements.

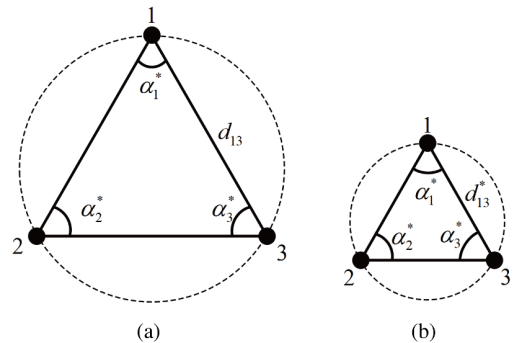


Figure 2 Uncertainty in formation scale based solely on angle control.

formation by controlling the distance between any two agents while maintaining angle constraints. In this case, we specifically choose the distance d_{13} between agent #1 and agent #3.

As shown in Figure 3(a), d_{13} will change while agent #1 and agent #2 remain stationary, and agent #3 moves along the straight line where the angle bisector of $\angle 132$ lies. Then, in Figure 3(b), let us take the formation reduction as an example. Agent #3 moves inward along the angle bisector, causing d_{13} to decrease. However, to ensure the constraints of the desired interior angles of the triangle, agent #1 and agent #2 will also move along the bisectors of their respective angles. As a result, the scale of the entire agent formation is reduced. To obtain d_{13} , we make the following assumption.

Assumption 4 Agent #3 can measure the distances between itself and the other agents. The other agents do not need to measure distance information.

Problem 1 Consider three agents with double-integrator dynamics described by eq. (1). The objective is to design a control law u_i for each agent i to facilitate the expansion and contraction of the formation while ensuring its stability. Then, extend the results to n agents. The stability of the entire agent system during formation maneuvers can be described as follows:

$$\begin{aligned} \lim_{t \rightarrow +\infty} (\alpha_i(t) - \alpha_i^*) &= 0, \\ \lim_{t \rightarrow +\infty} (\mathbf{v}_i(t) - \mathbf{v}_c^*) &= \mathbf{0}, \\ \lim_{t \rightarrow +\infty} (d_{ij}(t) - d_{ij}^*) &= 0, \end{aligned} \quad (5)$$

where α_i^* determines the desired formation shape, $\mathbf{v}_c^*$ determines the desired velocity of the formation, and d_{ij}^* denotes the desired distance between the agent i and j .

3 Main results

In this section, we introduce the agent formation control law and the stability analysis of the system. First, we present the formation control law for triangle configurations. Next, we extend it to the formation control of n agents using angle

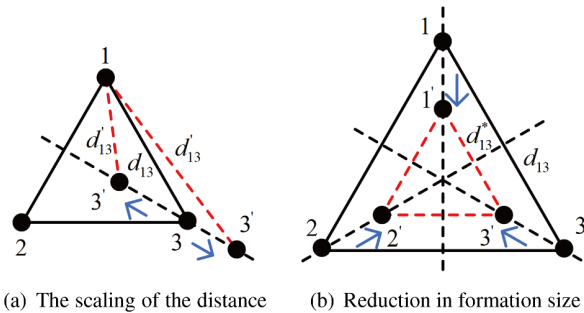


Figure 3 (Color online) The triangular formation motion strategy based on distance and angle.

rigidity. Finally, we discuss a construction scheme for the square formation.

3.1 Triangle formation control

We first consider the desired triangle formation and scaling. The formation control laws for agent #1 to #3 are proposed as follows:

$$\begin{aligned} u_1 &= \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_1 - \mathbf{v}_c^*) - k_2 (\alpha_1 - \alpha_1^*) (\mathbf{z}_{12} + \mathbf{z}_{13}), \\ u_2 &= \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_2 - \mathbf{v}_c^*) - k_2 (\alpha_2 - \alpha_2^*) (\mathbf{z}_{21} + \mathbf{z}_{23}), \\ u_3 &= \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_3 - \mathbf{v}_c^*) - k_2 (\alpha_3 - \alpha_3^*) (\mathbf{z}_{31} + \mathbf{z}_{32}) \\ &\quad + k_3 (d_{13} - d_{13}^*) (\mathbf{z}_{31} + \mathbf{z}_{32}), \end{aligned} \quad (6)$$

where $\dot{\mathbf{v}}_c^* \in \mathbb{R}^2$ is the desired acceleration, α_i^* is the desired interior angle of the desired triangle configuration, d_{13}^* is the desired distance between agent #1 and agent #3, and k_1 , k_2 and k_3 are positive scalar gains.

Before presenting the stability analysis, we first provide a lemma.

Lemma 1 For a matrix M of the form

$$M = \begin{bmatrix} \mathbf{0}_{n \times n} & \mathbf{I}_n \\ \mathbf{H}_{n \times n} & -k\mathbf{I}_n \end{bmatrix},$$

where $\mathbf{I}$ is the identity matrix and k is a positive constant. If the submatrix $\mathbf{H}$ is Hurwitz and the eigenvalues λ of $\mathbf{H}$ satisfy the inequality $(k^2 \operatorname{Re}[\lambda] + \operatorname{Im}^2[\lambda]) < 0$, then the matrix M is also Hurwitz.

Proof. The characteristic polynomial of M can be expressed as

$$\begin{aligned} |\lambda \mathbf{I}_{2n} - M| &= \begin{vmatrix} \lambda \mathbf{I}_n & -\mathbf{I}_n \\ -\mathbf{H} & (\lambda + k)\mathbf{I}_n \end{vmatrix} \\ &= |(\lambda + k)\mathbf{I}_n| \left| \lambda \mathbf{I}_n - \frac{(-\mathbf{I}_n)(-\mathbf{H})}{\lambda + k} \right| \\ &= |\lambda(\lambda + k)\mathbf{I}_n - \mathbf{H}|. \end{aligned} \quad (7)$$

We assume that $\lambda_1, \lambda_2, \dots, \lambda_n$ are the n eigenvalues of $\mathbf{H}$. For any $\mathbf{H}$, there always exists an $n \times n$ unitary matrix $\mathbf{U}$ such that

$$\mathbf{H} = \mathbf{U} \begin{bmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \mathbf{U}^{-1}. \quad (8)$$

Substituting eq. (8) into $|\lambda(\lambda + k)\mathbf{I}_n - \mathbf{H}|$, one has

$$\begin{aligned} |\lambda(\lambda + k)\mathbf{I}_n - \mathbf{H}| &= (\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2) \cdots (\lambda^2 + k\lambda - \lambda_n). \end{aligned} \quad (9)$$

For any real eigenvalue $\lambda_i, i \in \{1, 2, \dots, n\}$, since $\mathbf{H}$ is Hurwitz, λ_i is negative. Therefore, the roots of equation $(\lambda^2 + k\lambda - \lambda_i) = 0$ have negative real parts.

For any pair of complex conjugate eigenvalues, assuming λ_1 and λ_2 are a pair of complex conjugate eigenvalues of $\mathbf{H}$, the polynomial $(\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2)$ can be expressed as follows:

$$\begin{aligned} & (\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2) \\ &= \lambda^4 + 2k\lambda^3 + [k^2 - (\lambda_1 + \lambda_2)]\lambda^2 \\ & \quad - k(\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2. \end{aligned} \quad (10)$$

Then we have the Hurwitz determinant

$$\mathbf{H}_1 = \begin{vmatrix} a_3 & a_1 & 0 & 0 \\ a_4 & a_2 & a_0 & 0 \\ 0 & a_3 & a_1 & 0 \\ 0 & a_4 & a_2 & a_0 \end{vmatrix}, \quad (11)$$

where

$$\begin{aligned} a_4 &= 1, \\ a_3 &= 2k, \\ a_2 &= k^2 - (\lambda_1 + \lambda_2), \\ a_1 &= -k(\lambda_1 + \lambda_2), \\ a_0 &= \lambda_1\lambda_2. \end{aligned} \quad (12)$$

Since $k > 0$ and $\text{Re}[\lambda_j] < 0, j \in \{1, 2\}$, one has

$$\lambda_1 + \lambda_2 = 2\text{Re}[\lambda_j] < 0, \quad (13)$$

$$\lambda_1\lambda_2 = \text{Re}^2[\lambda_j] + \text{Im}^2[\lambda_j] > 0, \quad (14)$$

where we obtained a_4, a_3, a_2, a_1 and a_0 are all positive. Now, in view of the various orders of leading principal minor determinants of $\mathbf{H}_1$, we have

$$\Delta_1 = a_3 = 2k > 0, \quad (15)$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} a_3 & a_1 \\ a_4 & a_2 \end{vmatrix} \\ &= 2k[k^2 - (\lambda_1 + \lambda_2)] + k(\lambda_1 + \lambda_2) \\ &= 2k^3 - 2k\text{Re}[\lambda_j] > 0, \end{aligned} \quad (16)$$

$$\begin{aligned} \Delta_3 &= a_1\Delta_2 - a_0 \begin{vmatrix} a_3 & a_1 \\ 0 & a_3 \end{vmatrix} \\ &= -2k\text{Re}[\lambda_j](2k^3 - 2k\text{Re}[\lambda_j]) - 4k^2\lambda_1\lambda_2 \\ &= -4k^2(k^2\text{Re}[\lambda_j] + \text{Im}^2[\lambda_j]), \end{aligned} \quad (17)$$

$$\Delta_4 = a_0\Delta_3. \quad (18)$$

In view of eq. (17), the roots of the quartic polynomial in eq. (10) have negative real parts if and only if

$(k^2\text{Re}[\lambda_j] + \text{Im}^2[\lambda_j]) < 0, j \in \{1, 2\}$. As a result, the roots of the polynomial (9) have negative real parts if and only if $(k^2\text{Re}[\lambda_i] + \text{Im}^2[\lambda_i]) < 0, i \in \{1, 2, \dots, n\}$.

The lemma has been proved.

The theorem for triangular formation goes as follows.

Theorem 1 Consider a group of three agents with double-integrator dynamics given by eq. (1) under Assumptions 1–4. If the initial error $d_{ij}(0) - d_{ij}^*, \alpha_i(0) - \alpha_i^*$ and $\mathbf{v}_i(0) - \mathbf{v}_c^*$, for $i = 1, 2, 3, j \in \mathcal{N}_i$, are near the equilibrium point, the formation control law (6) ensures that $d_{ij}(t) \rightarrow d_{ij}^*, \alpha_i(t) \rightarrow \alpha_i^*$ and $\mathbf{v}_i(t) \rightarrow \mathbf{v}_c^*$ asymptotically as $t \rightarrow +\infty$.

Proof. At first, since $\frac{d(\cos \alpha_i)}{dt} = -(\sin \alpha_i)\dot{\alpha}_i$ and from eq. (3), the interior angle dynamics of each agent can be given by

$$\begin{aligned} \dot{\alpha}_1 &= -\frac{\mathbf{z}_{13}^\top \mathbf{P}_{z_{12}}(\mathbf{v}_2 - \mathbf{v}_1)}{d_{12} \sin \alpha_1} - \frac{\mathbf{z}_{12}^\top \mathbf{P}_{z_{13}}(\mathbf{v}_3 - \mathbf{v}_1)}{d_{13} \sin \alpha_1}, \\ \dot{\alpha}_2 &= -\frac{\mathbf{z}_{23}^\top \mathbf{P}_{z_{21}}(\mathbf{v}_1 - \mathbf{v}_2)}{d_{21} \sin \alpha_2} - \frac{\mathbf{z}_{21}^\top \mathbf{P}_{z_{23}}(\mathbf{v}_3 - \mathbf{v}_2)}{d_{23} \sin \alpha_2}, \\ \dot{\alpha}_3 &= -\frac{\mathbf{z}_{32}^\top \mathbf{P}_{z_{31}}(\mathbf{v}_1 - \mathbf{v}_3)}{d_{31} \sin \alpha_3} - \frac{\mathbf{z}_{31}^\top \mathbf{P}_{z_{32}}(\mathbf{v}_2 - \mathbf{v}_3)}{d_{32} \sin \alpha_3}, \end{aligned} \quad (19)$$

where $\mathbf{P}_{z_{ij}} = \mathbf{I}_2 - \mathbf{z}_{ij}\mathbf{z}_{ij}^\top \in \mathbb{R}^{2 \times 2}$.

Define $e_i = \alpha_i - \alpha_i^*$, then we have the angle error dynamics as follows:

$$\dot{e}_i = -N_{jik}(\mathbf{v}_k - \mathbf{v}_j) - N_{kij}(\mathbf{v}_j - \mathbf{v}_k), \quad (20)$$

where $N_{jik} = \mathbf{z}_{ij}^\top \frac{\mathbf{P}_{z_{ik}}}{d_{ik} \sin \alpha_{jik}} \in \mathbb{R}^{1 \times 2}, \forall j, i, k \in \{1, 2, 3\}$.

Second, the velocity error dynamics of $\mathbf{v}_i - \mathbf{v}_c^*$ can be derived from eq. (6); one has

$$\dot{\mathbf{v}}_i - \dot{\mathbf{v}}_c^* = -k_1(\mathbf{v}_i - \mathbf{v}_c^*) - k_2(\alpha_i - \alpha_i^*)(\mathbf{z}_{i(i-1)} + \mathbf{z}_{i(i+1)}). \quad (21)$$

Particularly, since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, the agent #3's velocity error dynamics can be rewritten as

$$\begin{aligned} \dot{\mathbf{v}}_3 - \dot{\mathbf{v}}_c^* &= -k_1(\mathbf{v}_3 - \mathbf{v}_c^*) + k_3(d_{13} - d_{13}^*)(\mathbf{z}_{31} + \mathbf{z}_{32}) \\ & \quad - k_2(\alpha_1^* - \alpha_1 + \alpha_2^* - \alpha_2)(\mathbf{z}_{31} + \mathbf{z}_{32}). \end{aligned} \quad (22)$$

Finally, we analyze the dynamic error of relative distance. Since $d_{13}^2 = \mathbf{p}_{13}^\top \mathbf{p}_{13}$, we have

$$2d_{13}\dot{d}_{13} = \dot{\mathbf{p}}_{13}^\top \mathbf{p}_{13} + \mathbf{p}_{13}^\top \dot{\mathbf{p}}_{13}, \quad (23)$$

then

$$\dot{d}_{13} = \mathbf{z}_{13}^\top [(\mathbf{v}_3 - \mathbf{v}_c^*) - (\mathbf{v}_1 - \mathbf{v}_c^*)]. \quad (24)$$

We define the system state variables

$$\mathbf{X} = [d_{13} - d_{13}^*, e_1, e_2, \mathbf{v}_1^\top - \mathbf{v}_c^{*\top}, \mathbf{v}_2^\top - \mathbf{v}_c^{*\top}, \mathbf{v}_3^\top - \mathbf{v}_c^{*\top}]^\top, \quad (25)$$

where $X \in \mathbb{R}^9$. According to eqs. (19)–(24), we have the closed loop system as follows:

$$\dot{X} = AX = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{R}_1 \\ \mathbf{R}_2 & -k_1 \mathbf{I}_6 \end{bmatrix} X, \quad (26)$$

where $\mathbf{R}_1 \in \mathbb{R}^{3 \times 6}$, $\mathbf{R}_2 \in \mathbb{R}^{6 \times 3}$, and

$$\mathbf{R}_1 = \begin{bmatrix} -\mathbf{z}_{13}^\top & \mathbf{0} & \mathbf{z}_{13}^\top \\ N_{312} + N_{213} & -N_{312} & -N_{213} \\ -N_{321} & N_{32} + N_{123} & -N_{123} \end{bmatrix}, \quad (27)$$

$$\mathbf{R}_2 = \begin{bmatrix} \mathbf{0} & -k_2(\mathbf{z}_{12} + \mathbf{z}_{13}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -k_2(\mathbf{z}_{21} + \mathbf{z}_{23}) \\ k_3(\mathbf{z}_{31} + \mathbf{z}_{32}) & k_2(\mathbf{z}_{31} + \mathbf{z}_{32}) & k_2(\mathbf{z}_{31} + \mathbf{z}_{32}) \end{bmatrix}. \quad (28)$$

Then, we linearize $\mathbf{R}_1$ and $\mathbf{R}_2$ near the equilibrium $X = \mathbf{0}$ to analyze the stability of the system:

$$\dot{X} = \bar{A}X = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \bar{\mathbf{R}}_1 \\ \bar{\mathbf{R}}_2 & -k_1 \mathbf{I}_6 \end{bmatrix} X, \quad (29)$$

where $\bar{A}$ is the linearization of the system matrix near the origin. For convenience, the symbol $(\bar{\cdot})$ denotes that the variable is assigned a value at $X = \mathbf{0}$ below. Next, we prove that system (29) is stable by analyzing the eigenvalues of $\bar{A}$, and prove that $\bar{A}$ is Hurwitz. Let λ be an eigenvalue of $\bar{A}$, then we have characteristic equation as follows:

$$\begin{aligned} |\lambda \mathbf{I}_9 - \bar{A}| &= \begin{vmatrix} \lambda \mathbf{I}_3 & -\bar{\mathbf{R}}_1 \\ -\bar{\mathbf{R}}_2 & (\lambda + k_1) \mathbf{I}_6 \end{vmatrix} \\ &= (\lambda + k_1)^3 |\lambda(\lambda + k_1) \mathbf{I}_3 - \bar{\mathbf{R}}_1 \bar{\mathbf{R}}_2|. \end{aligned} \quad (30)$$

From eq. (30), we find that $-k_1$ is the triple root of $\bar{A}$, also, $-k_1 < 0$. Then, we analyze the eigenvalues of matrix $\bar{\mathbf{R}}_1 \bar{\mathbf{R}}_2$. Let $\bar{\mathbf{G}}_1 = \bar{\mathbf{R}}_1 \bar{\mathbf{R}}_2$, one has

$$\bar{\mathbf{G}}_1 = \begin{bmatrix} \bar{\mathbf{G}}_{11} & \bar{\mathbf{G}}_{12} & \bar{\mathbf{G}}_{13} \end{bmatrix}, \quad (31)$$

where $\bar{\mathbf{G}}_{1i} \in \mathbb{R}^{3 \times 1}$ is the i th column of $\bar{\mathbf{G}}_1$, $i \in \{1, 2, 3\}$. By using $\mathbf{P}_{z_{ij}} \mathbf{z}_{ij} = 0$, we have $N_{jik} \mathbf{z}_{ik} = 0$ and $N_{jik} \mathbf{z}_{ki} = 0$. Then we can simplify to obtain

$$\bar{\mathbf{G}}_{11} = \begin{bmatrix} k_3 \mathbf{z}_{13}^\top (\mathbf{z}_{31} + \mathbf{z}_{32}) \\ -k_3 N_{213} \mathbf{z}_{32} \\ -k_3 N_{123} \mathbf{z}_{31} \end{bmatrix}, \quad (32)$$

$$\bar{\mathbf{G}}_{12} = \begin{bmatrix} k_2 \mathbf{z}_{13}^\top (\mathbf{z}_{12} + \mathbf{z}_{32}) \\ -k_2 (N_{312} \mathbf{z}_{13} + N_{213} \mathbf{z}_{12} + N_{213} \mathbf{z}_{32}) \\ k_2 (N_{321} \mathbf{z}_{13} - N_{123} \mathbf{z}_{31}) \end{bmatrix}, \quad (33)$$

$$\bar{\mathbf{G}}_{13} = \begin{bmatrix} k_2 \mathbf{z}_{13}^\top (\mathbf{z}_{31} + \mathbf{z}_{32}) \\ k_2 (N_{312} \mathbf{z}_{23} - N_{213} \mathbf{z}_{32}) \\ -k_2 (N_{321} \mathbf{z}_{23} + N_{123} \mathbf{z}_{21} + N_{123} \mathbf{z}_{31}) \end{bmatrix}. \quad (34)$$

Simplifying further by using $\cos \alpha_{jik} = \mathbf{z}_{ij}^\top \mathbf{z}_{ik}$, one has

$$\bar{\mathbf{G}}_1 = \begin{bmatrix} -k_3(1 + \cos \alpha_3) & -k_2(\cos \alpha_3 - \cos \alpha_1) & -k_2(1 + \cos \alpha_3) \\ -k_3 f_{13} & -k_2(g_1 + f_{13}) & -k_2(f_{13} - f_{12}) \\ -k_3 f_{23} & -k_2(f_{23} - f_{21}) & -k_2(g_2 + f_{23}) \end{bmatrix}, \quad (35)$$

where $f_{ij} = \frac{\sin \alpha_j}{d_{ij}}$ and $g_i = (\sin \alpha_i) \left(\frac{1}{d_{ij}} + \frac{1}{d_{ik}} \right)$, $i, j, k \in \{1, 2, 3\}$, and $j, k \in \mathcal{N}_i$.

Let $p = 1 + \cos \alpha_3$ and $q = \cos \alpha_3 - \cos \alpha_1$, eq. (35) can be rewritten as

$$\bar{\mathbf{G}}_1 = \begin{bmatrix} -k_3 p & -k_2 q & -k_2 p \\ -k_3 f_{13} & -k_2(g_1 + f_{13}) & -k_2(f_{13} - f_{12}) \\ -k_3 f_{23} & -k_2(f_{23} - f_{21}) & -k_2(g_2 + f_{23}) \end{bmatrix}. \quad (36)$$

Then, let λ_1 be an eigenvalue of $\bar{\mathbf{G}}_1$, one has

$$\begin{aligned} |\lambda_1 \mathbf{I} - \bar{\mathbf{G}}_1| &= \begin{vmatrix} \lambda_1 + k_3 p & k_2 q & k_2 p \\ k_3 f_{13} & \lambda_1 + k_2(g_1 + f_{13}) & k_2(f_{13} - f_{12}) \\ k_3 f_{23} & k_2(f_{23} - f_{21}) & \lambda_1 + k_2(g_2 + f_{23}) \end{vmatrix} \\ &= a_3 \lambda_1^3 + a_2 \lambda_1^2 + a_1 \lambda_1 + a_0, \end{aligned} \quad (37)$$

where

$$\begin{aligned} a_3 &= 1, \\ a_2 &= k_2 f_{13} + k_2 f_{23} + k_2 g_1 + k_2 g_2 + k_3 p, \\ a_1 &= k_2^2 f_{13} f_{21} + k_2^2 f_{12} f_{23} + k_2^2 f_{13} g_2 \\ &\quad + k_2^2 f_{23} g_1 + k_2^2 (g_1 g_2 - f_{12} f_{21}) \\ &\quad + k_2 k_3 f_{13} (p - q) + k_2 k_3 g_1 p + k_2 k_3 g_2 p, \\ a_0 &= k_2^2 k_3 [f_{12} f_{23} (p - q) + f_{13} g_2 (p - q) \\ &\quad + (g_1 g_2 - f_{12} f_{21}) p]. \end{aligned} \quad (38)$$

Now, we focus on the coefficients of the roots of the cubic polynomial of eq. (37) by using the condition of Routh-Hurwitz criteria. Considering $p > 0$, $p - q > 0$, and $g_1 g_2 - f_{12} f_{21} > 0$, we can get that a_3 , a_2 , a_1 and a_0 are all positive. Then, we obtained the Hurwitz determinant

$$\mathbf{H} = \begin{bmatrix} a_2 & a_3 & 0 \\ a_0 & a_1 & a_2 \\ 0 & 0 & a_0 \end{bmatrix}, \quad (39)$$

one has

$$\Delta_1 = a_2 > 0, \quad (40)$$

$$\Delta_2 = \begin{vmatrix} a_2 & a_3 \\ a_0 & a_1 \end{vmatrix} = a_2 a_1 - a_3 a_0 > 0, \quad (41)$$

$$\Delta_3 = a_0 \Delta_2 > 0, \quad (42)$$

where inequality (41) is calculated in Appendix. By checking the above Routh-Hurwitz conditions, we obtained the characteristic roots of eq. (37) all have negative real parts. Therefore, $\bar{G}_1$ is Hurwitz.

In summary, $\bar{A}$ is Hurwitz under the Lemma, and it can be concluded that the agent formation system is exponentially stable when the initial errors are located near the equilibrium point.

Remark 1 It should be pointed out that, for the proposed control law, the number of distance measurements is reduced. When it is necessary to control expansion and contraction to avoid obstacles, it is not necessary for each agent to measure distance information, nor is it necessary for a large number of agents to measure distance information.

Remark 2 In view of Figure 4, note that the local bearings of each agent can be misaligned. Even if the agents lose their global sense of direction in certain complex environments, it will not affect the completion of the formation. The local bearing misalignment does not affect the interior angle α_i and the distance d_{ij} measurements. Regardless of the bearing orientation, the agents' formation motion is always toward the angular bisector $\mathbf{z}_{ij} + \mathbf{z}_{ik}$ of the interior angle α_i , and the direction of $\mathbf{z}_{ij} + \mathbf{z}_{ik}$ is constant.

3.2 Polygonal formation

In this subsection, we will give the polygonal formation control law of n agents. We consider a formation of n agents and expand the team scale of the formation through the method shown in Figure 5. For agent i , $i > 3$, each agent needs to measure two angles $\alpha_{j_1 i j_2}^*$ and $\alpha_{j_2 i j_3}^*$ relative to three neighbor agents $j_1, j_2, j_3 < i$, $j_1, j_2, j_3 \in \mathcal{N}_i$. We have an angle set $\mathcal{A} = \{(j_1, i, j_2), (j_2, i, j_3)\}$.

Assumption 5 When using the above method to expand the team scale of the formation, we assume that the desired positions of agent $i > 3$, agent j_1 , agent j_2 and agent j_3 are not in a circle, $\alpha_{j_1 i j_2}^*$ and $\alpha_{j_2 i j_3}^*$ are not zero or π .

Remark 3 When Assumption 5 holds and the formation configuration is constructed in the manner depicted in Figure 5, the formation is angularly rigid, and the analysis of angle rigidity can be found in ref. [26].

To solve the problem proposed in Section 2, the formation control laws are designed as

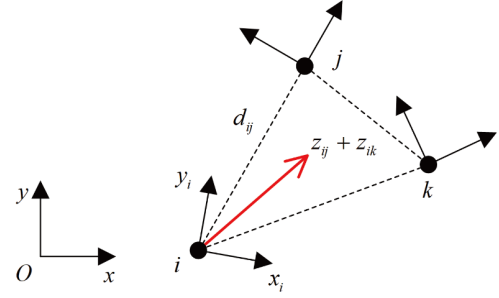


Figure 4 (Color online) The local bearings are not aligned.

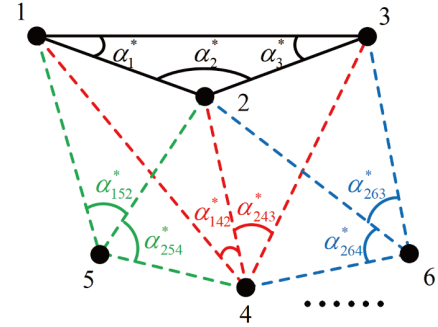


Figure 5 (Color online) Expansion of formation team scale.

$$\begin{aligned} \mathbf{u}_i = & \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_i - \mathbf{v}_c^*) + k_3 (d_{13} - d_{13}^*) (\mathbf{z}_{ij} + \mathbf{z}_{ik}) \\ & - k_2 \sum_{(j,i,k) \in \mathcal{A}} (\alpha_{jik} - \alpha_{jik}^*) (\mathbf{z}_{ij} + \mathbf{z}_{ik}), \end{aligned} \quad (43)$$

where k_1 and k_2 are positive scalar gains, k_3 is a positive constant if $i = 3$, and $k_3 = 0$ otherwise.

Theorem 2 Consider a group of n agents with double-integrator dynamics given by eq. (1) under Assumptions 1 – 5. If the initial error $d_{ij}(0) - d_{ij}^*$, $\alpha_i(0) - \alpha_i^*$ and $\mathbf{v}_i(0) - \mathbf{v}_c^*$ are near the equilibrium point, the formation control law (43) ensures that $d_{ij}(t) \rightarrow d_{ij}^*$, $\alpha_i(t) \rightarrow \alpha_i^*$ and $\mathbf{v}_i(t) \rightarrow \mathbf{v}_c^*$ asymptotically as $t \rightarrow +\infty$.

Proof. The system state variables are defined as

$$\begin{aligned} \mathbf{X} = & [d_{13} - d_{13}^*, e_1, e_2, e_{41}, e_{42}, \dots, e_{n1}, e_{n2}, \mathbf{v}_1^\top - \mathbf{v}_c^{*\top}, \\ & \mathbf{v}_2^\top - \mathbf{v}_c^{*\top}, \mathbf{v}_3^\top - \mathbf{v}_c^{*\top}, \dots, \mathbf{v}_n^\top - \mathbf{v}_c^{*\top}]^\top \in \mathbb{R}^{4n-3}. \end{aligned} \quad (44)$$

According to the previous foreshadowing, we directly give the closed-loop dynamics

$$\dot{\mathbf{X}} = \mathbf{B}\mathbf{X} = \begin{bmatrix} \mathbf{0}_{(2n-3) \times (2n-3)} & \mathbf{S}_1 \\ \mathbf{S}_2 & -k_1 \mathbf{I}_{2n} \end{bmatrix} \mathbf{X}, \quad (45)$$

where $\mathbf{S}_1 \in \mathbb{R}^{(2n-3) \times 2n}$, $\mathbf{S}_2 \in \mathbb{R}^{2n \times (2n-3)}$, and

$\mathbf{S}_1$

$$= \begin{bmatrix} -z_{13}^\top & \mathbf{0} & z_{13}^\top & \mathbf{0} & \cdots & \mathbf{0} \\ N_{312} + N_{213} & -N_{312} & -N_{213} & \mathbf{0} & \cdots & \mathbf{0} \\ -N_{321} & N_{32} + N_{123} & -N_{123} & \mathbf{0} & \cdots & \mathbf{0} \\ -N_{241} & -N_{142} & \mathbf{0} & N_{241} + N_{142} & \cdots & \mathbf{0} \\ \mathbf{0} & -N_{342} & -N_{243} & N_{342} + N_{243} & \cdots & \mathbf{0} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}, \quad (46)$$

$$S_2 = \begin{bmatrix} \mathbf{0} & -k_2(z_{12} + z_{13}) & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -k_2(z_{21} + z_{33}) & \cdots & \mathbf{0} \\ k_3(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) & \cdots & \mathbf{0} \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}. \quad (47)$$

Then, we linearize S_1 and S_2 near the equilibrium $X = \mathbf{0}$ to analyze the stability of the system, one has

$$\dot{X} = \bar{B}X = \begin{bmatrix} \mathbf{0}_{(2n-3) \times (2n-3)} & \bar{S}_1 \\ \bar{S}_2 & -k_1 I_{2n} \end{bmatrix} X. \quad (48)$$

We have the characteristic equation as follows:

$$|\lambda I_{4n-3} - \bar{B}| = \begin{vmatrix} \lambda I_{2n-3} & -\bar{S}_1 \\ -\bar{S}_2 & (\lambda + k_1) I_{2n} \end{vmatrix} = (\lambda + k_1)^3 |\lambda(\lambda + k_1) I_{2n-3} - \bar{S}_1 \bar{S}_2|, \quad (49)$$

and in eq. (49),

$$\bar{S}_1 \bar{S}_2 = \begin{bmatrix} \bar{G}_1 & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ * & \bar{G}_4 & \mathbf{0} & \cdots & \mathbf{0} \\ * & * & \bar{G}_5 & \cdots & \mathbf{0} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \cdots & \bar{G}_n \end{bmatrix}, \quad (50)$$

where $\bar{G}_1 \in \mathbb{R}^{3 \times 3}$, $\bar{G}_4, \bar{G}_5, \dots, \bar{G}_n \in \mathbb{R}^{2 \times 2}$, and matrix $\bar{S}_1 \bar{S}_2$ is a block lower triangular matrix. For $\bar{G}_i, i > 3$

$$\bar{G}_i = k_2 \bar{F}_i = k_2 \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad (51)$$

where

$$\begin{aligned} a &= -k_2 (N_{j_1 i j_2} + N_{j_2 i j_1}) (z_{i j_1} + z_{i j_2}), \\ b &= -k_2 (N_{j_1 i j_2} + N_{j_2 i j_1}) (z_{i j_2} + z_{i j_3}), \\ c &= -k_2 (N_{j_2 i j_3} + N_{j_3 i j_2}) (z_{i j_1} + z_{i j_2}), \\ d &= -k_2 (N_{j_2 i j_3} + N_{j_3 i j_2}) (z_{i j_2} + z_{i j_3}). \end{aligned}$$

Let λ_1 and λ_2 be the two eigenvalues of $\bar{F}_i$ respectively.

By using

$$\begin{cases} \lambda_1 + \lambda_2 = \text{tr}(\bar{F}_i) < 0, \\ \lambda_1 \lambda_2 = \det(\bar{F}_i) > 0, \end{cases} \quad (52)$$

we have $\bar{G}_i$ is Hurwitz. Then, substituting eq. (50) into eq. (49), the characteristic polynomial can be rewritten as

$$|\lambda I_{4n-3} - \bar{B}| = (\lambda + k_1)^3 \left| \lambda(\lambda + k_1) I_3 - \bar{G}_1 \right| \prod_{i=4}^n \left| \lambda(\lambda + k_1) I_2 - \bar{G}_i \right|. \quad (53)$$

Since $\bar{G}_i$ is Hurwitz, we give the conclusion directly that Theorem 2 is proved under Lemma 1.

Remark 4 It should be pointed out that there is angle rigidity under Assumption 5. However, the formation scale is not rigid. Below we explain why only the change in distance between a pair of agents can determine the scale of the entire formation.

As shown in Figure 2, according to the law of sines

$$\frac{d_{23}}{\sin \alpha_1^*} = \frac{d_{13}}{\sin \alpha_2^*} = \frac{d_{12}}{\sin \alpha_3^*} = 2R, \quad (54)$$

where R denotes the radius of the circumcircle of $\triangle 123$, when d_{13} shrinks $k = \frac{d_{13}}{d_{13}^*}$ times, the radius of the circumcircle also shrinks k times.

Then, since a chord and an angle can uniquely determine a circumcircle, when the angle is determined, the radius of the circumscribed circle is only related to the length of the chord. For example, the chord 12 and α_{142}^* is given in Figure 6(a) and (b). As a result, the scale of a formation of four agents can be completely determined by the distance on either side of the first triangle $\triangle 123$.

According to mathematical induction, it can be derived to n agents.

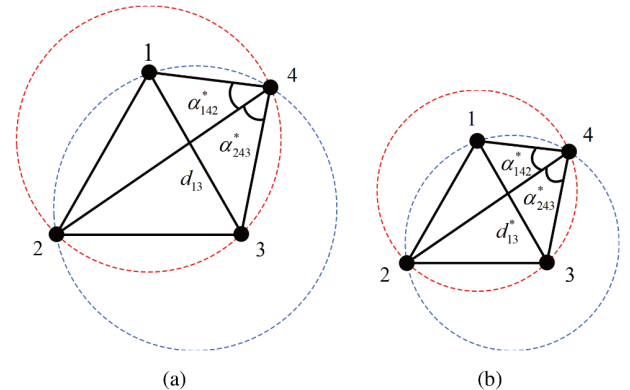


Figure 6 (Color online) The uniqueness of a formation.

3.3 Square formation control

In this subsection, we discuss the control for square formation with four agents on a circle.

As shown in Figure 6(a), the desired formations of agent #4 and the first three agents are not on the same circle, which satisfies Assumption 5. This constitutes a condition for the validity of the angle rigidity theory. Therefore, the desired position of agent #4 is unique under the constraints of α_{142}^* and α_{243}^* . Here, we relax the condition. When the four agents are on a circle, that is, when the red circle and the blue circle coincide in Figure 6, the formation configuration satisfies the expected angle and distance; however, the configuration is not unique as shown in Figure 7. Agent #4 measures the LOS angle to agent #1, #2 and #3 respectively, forming angle constraints, where $\alpha_{142}^* = \pi/4$ and $\alpha_{243}^* = \pi/4$ in Figure 7(a). Obviously, the position of agent #4 is not unique, because agent #4 can move arbitrarily along the arc 143 as shown in Figure 7(b).

In the configuration depicted in Figure 8(a), where agent #3 measures only d_{32} , d_{34} may be ambiguous. To solve the above problems, we can change the constraints as shown in Figure 8(b). Agent #1, agent #2, agent #3 and agent #4 each measure the interior angle α_1 , α_2 , α_3 , and α_4 with its

neighbors. Then agent #3 measures the distance d_{32} and d_{34} . Therefore, the problem can be formulated as follows:

$$\begin{aligned} \lim_{t \rightarrow +\infty} e_i &= \lim_{t \rightarrow +\infty} (\alpha_i - \pi/2) = 0, \\ \lim_{t \rightarrow +\infty} (d_{32} - d_{32}^*) &= 0, \\ \lim_{t \rightarrow +\infty} (d_{34} - d_{34}^*) &= 0. \end{aligned} \quad (55)$$

The square formation control laws for agent #1 to #4 are proposed to be

$$\begin{aligned} \mathbf{u}_i &= \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_i - \mathbf{v}_c^*) - k_2 (\alpha_i - \alpha_i^*) (\mathbf{z}_{i(i-1)} + \mathbf{z}_{i(i+1)}), \\ i &\in \{1, 2, 4\}, \\ \mathbf{u}_3 &= \dot{\mathbf{v}}_c^* - k_1 (\mathbf{v}_3 - \mathbf{v}_c^*) - k_2 (\alpha_3 - \alpha_3^*) (\mathbf{z}_{32} + \mathbf{z}_{34}) \\ &\quad + k_3 (d_{32} - d_{32}^*) \mathbf{z}_{32} + k_3 (d_{34} - d_{34}^*) \mathbf{z}_{34}. \end{aligned} \quad (56)$$

Theorem 3 Consider a group of four agents with double-integrator dynamics given by eq. (1) under Assumptions 1–4. If the initial error $d_{ij}(0) - d_{ij}^*$, $\alpha_i(0) - \alpha_i^*$, and $\mathbf{v}_i(0) - \mathbf{v}_c^*$, for $i = 1, 2, 3, 4$, $j \in \mathcal{N}_i$, are near the equilibrium point, the square formation control law (56) ensures that $d_{ij}(t) \rightarrow d_{ij}^*$, $\alpha_i(t) \rightarrow \alpha_i^*$, and $\mathbf{v}_i(t) \rightarrow \mathbf{v}_c^*$ asymptotically as $t \rightarrow +\infty$.

$$\begin{aligned} &\begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} \\ &= \begin{bmatrix} -k_3 & -k_3 \cos \alpha_3 & 0 & k_2 (1 + \cos \alpha_2) & k_2 (1 + \cos \alpha_3) \\ -k_3 \cos \alpha_3 & -k_3 & -k_2 (1 + \cos \alpha_4) & -k_2 (1 + \cos \alpha_4) & k_2 (\cos \alpha_3 - \cos \alpha_4) \\ 0 & 0 & -k_2 \left(\frac{\sin \alpha_1}{d_{12}} + \frac{\sin \alpha_1}{d_{14}} + \frac{\sin \alpha_4}{d_{14}} \right) & k_2 \left(\frac{\sin \alpha_2}{d_{12}} - \frac{\sin \alpha_4}{d_{14}} \right) & -k_2 \frac{\sin \alpha_4}{d_{14}} \\ 0 & -k_3 \frac{\sin \alpha_3}{d_{23}} & k_2 \frac{\sin \alpha_1}{d_{12}} & -k_2 \left(\frac{\sin \alpha_2}{d_{23}} + \frac{\sin \alpha_2}{d_{12}} \right) & k_2 \frac{\sin \alpha_3}{d_{23}} \\ k_3 \frac{\sin \alpha_3}{d_{34}} & k_3 \frac{\sin \alpha_3}{d_{23}} & -k_2 \frac{\sin \alpha_4}{d_{34}} & k_2 \left(\frac{\sin \alpha_2}{d_{23}} - \frac{\sin \alpha_4}{d_{34}} \right) & -k_2 \left(\frac{\sin \alpha_3}{d_{34}} + \frac{\sin \alpha_3}{d_{23}} + \frac{\sin \alpha_4}{d_{34}} \right) \end{bmatrix}. \end{aligned} \quad (57)$$

Proof. With the control law (56), the angle error dynamics of each agent can be given by

$$\dot{e}_i = -N_{jik} (\mathbf{v}_k - \mathbf{v}_j) - N_{kij} (\mathbf{v}_j - \mathbf{v}_k). \quad (58)$$

The velocity error dynamics are given by

$$\begin{aligned} \dot{\mathbf{v}}_i - \dot{\mathbf{v}}_c^* &= -k_1 (\mathbf{v}_i - \mathbf{v}_c^*) - k_2 (\alpha_i - \alpha_i^*) (\mathbf{z}_{i(i-1)} + \mathbf{z}_{i(i+1)}), \\ i &\in \{1, 2, 4\}, \\ \dot{\mathbf{v}}_3 - \dot{\mathbf{v}}_c^* &= -k_1 (\mathbf{v}_3 - \mathbf{v}_c^*) - k_2 (\alpha_3 - \alpha_3^*) (\mathbf{z}_{32} + \mathbf{z}_{34}) \\ &\quad + k_3 (d_{32} - d_{32}^*) \mathbf{z}_{32} + k_3 (d_{34} - d_{34}^*) \mathbf{z}_{34}. \end{aligned} \quad (59)$$

The distance error dynamics are given by

$$\begin{aligned} \dot{d}_{32} &= \mathbf{z}_{32}^\top [(\mathbf{v}_2 - \mathbf{v}_c^*) - (\mathbf{v}_3 - \mathbf{v}_c^*)], \\ \dot{d}_{34} &= \mathbf{z}_{34}^\top [(\mathbf{v}_4 - \mathbf{v}_c^*) - (\mathbf{v}_3 - \mathbf{v}_c^*)]. \end{aligned} \quad (60)$$

Since $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 2\pi$, let the system state variables be

$$\begin{aligned} \mathbf{X} &= [d_{32} - d_{32}^*, d_{34} - d_{34}^*, e_1, e_2, e_3, \mathbf{v}_1^\top - \mathbf{v}_c^{*\top}, \\ &\quad \mathbf{v}_2^\top - \mathbf{v}_c^{*\top}, \mathbf{v}_3^\top - \mathbf{v}_c^{*\top}, \mathbf{v}_4^\top - \mathbf{v}_c^{*\top}]^\top \in \mathbb{R}^{13}. \end{aligned} \quad (61)$$

Then we have the closed-loop system linearized at the

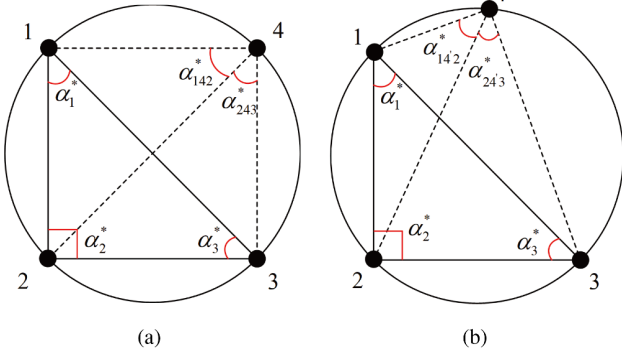


Figure 7 (Color online) Uncertainty of formation on a circle.

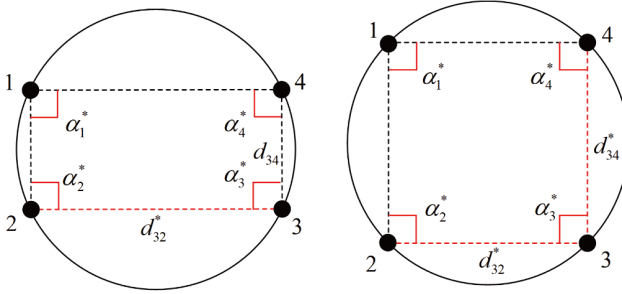


Figure 8 (Color online) Square formation construction.

equilibrium point as follows:

$$\dot{X} = \bar{C}X = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 3} & \bar{T}_1 \\ \mathbf{0}_{3 \times 2} & \mathbf{0}_{3 \times 3} & \bar{T}_2 \\ \bar{T}_3 & \bar{T}_4 & -k_1 \mathbf{I}_8 \end{bmatrix} X, \quad (62)$$

where $\bar{T}_1 \in \mathbb{R}^{2 \times 8}$, $\bar{T}_2 \in \mathbb{R}^{3 \times 8}$, $\bar{T}_3 \in \mathbb{R}^{8 \times 2}$, $\bar{T}_4 \in \mathbb{R}^{8 \times 3}$, and

$$\begin{aligned} \bar{T}_1 &= \begin{bmatrix} \mathbf{0} & \mathbf{z}_{32}^\top & -\mathbf{z}_{32}^\top & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{z}_{34}^\top & \mathbf{z}_{34}^\top \end{bmatrix}, \\ \bar{T}_2 &= \begin{bmatrix} N_{412} + N_{214} & -N_{412} & \mathbf{0} & -N_{214} \\ -N_{321} & N_{123} + N_{321} & -N_{123} & \mathbf{0} \\ \mathbf{0} & -N_{432} & N_{234} + N_{432} & -N_{234} \end{bmatrix}, \\ \bar{T}_3 &= \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \\ k_3 \mathbf{z}_{32} & k_3 \mathbf{z}_{34} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, \\ \bar{T}_4 &= \begin{bmatrix} -k_2 (\mathbf{z}_{12} + \mathbf{z}_{14}) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -k_2 (\mathbf{z}_{21} + \mathbf{z}_{23}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -k_2 (\mathbf{z}_{32} + \mathbf{z}_{34}) \\ k_2 (\mathbf{z}_{41} + \mathbf{z}_{43}) & k_2 (\mathbf{z}_{41} + \mathbf{z}_{43}) & k_2 (\mathbf{z}_{41} + \mathbf{z}_{43}) \end{bmatrix}. \end{aligned} \quad (63)$$

Thus, we can obtain the characteristic equation:

$$\begin{aligned} & |\lambda \mathbf{I}_{13} - \bar{C}| \\ &= \begin{vmatrix} \lambda \mathbf{I}_5 & -\begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \\ -\begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} & (\lambda + k_1) \mathbf{I}_8 \end{vmatrix} \\ &= (\lambda + k_1)^3 \left| \lambda (\lambda + k_1) \mathbf{I}_5 - \begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} \right|, \end{aligned} \quad (64)$$

where $\begin{bmatrix} \bar{T}_1 & \bar{T}_2 \end{bmatrix}^\top \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix}$ is given by eq. (57). Let $\mathbf{H} = \begin{bmatrix} \bar{T}_1 & \bar{T}_2 \end{bmatrix}^\top \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix}$. The determinant $|\lambda (\lambda + k_1) \mathbf{I}_5 - \mathbf{H}|$ is equivalent to the characteristic equation associated with the matrix $\begin{bmatrix} \mathbf{0} & \mathbf{I}_5 \\ \mathbf{H} & -k_1 \mathbf{I}_5 \end{bmatrix}$. Thus, from the lemma, if $\mathbf{H}$ is Hurwitz, we have $\bar{C}$ is Hurwitz so that the agent square formation system is exponentially stable.

Remark 5 Note that the movement direction of the convergence part of the distance error is changed in control law (56). Previously, in the control law (6), the movement direction of the dynamic error of the distance was the angle bisector $\mathbf{z}_{ij} + \mathbf{z}_{ik}$, that is, the weight of the dynamic error of the distance in $\mathbf{z}_{ij}$ and $\mathbf{z}_{ik}$ was equal. Here, the weights on $\mathbf{z}_{ij}$ and $\mathbf{z}_{ik}$ become the respective distance dynamic errors. Seen in Figure 9, the direction of movement becomes $e_{d_{ij}} \mathbf{z}_{ij} + e_{d_{ik}} \mathbf{z}_{ik}$, where $e_{d_{ij}} = d_{ij} - d_{ij}^*$.

4 Simulations

In this section, we use the simplified model of the quadrotor UAV to verify the validity of the theoretical results in Section 3. The low-level control of each quadrotor is determined by an outer position loop and an inner attitude loop [31]. The position model can be decoupled into two linear models, namely the horizontal channel model and the altitude channel model [32, Chapter 11]:

$$\begin{cases} \dot{\mathbf{p}}_h = \mathbf{v}_h, \\ \dot{\mathbf{v}}_h = -g \mathbf{A}_\psi \Theta_h, \end{cases} \quad (65)$$

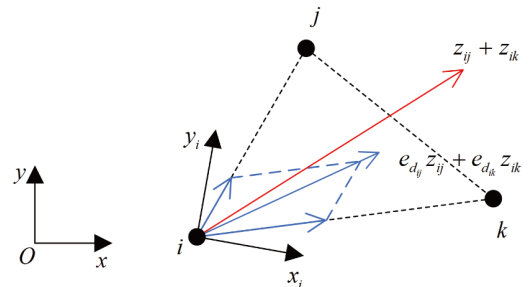


Figure 9 (Color online) Direction of motion caused by range dynamic error.

$$\begin{cases} \dot{p}_z = v_z, \\ \dot{v}_z = g - \frac{f}{m}, \end{cases} \quad (66)$$

where

$$\begin{aligned} \mathbf{p}_h &= \begin{bmatrix} p_x \\ p_y \end{bmatrix}, \mathbf{R}_\psi = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}, \\ \mathbf{A}_\psi &= \mathbf{R}_\psi \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \boldsymbol{\Theta}_h = \begin{bmatrix} \phi \\ \theta \end{bmatrix}. \end{aligned} \quad (67)$$

Here, $g \in \mathbb{R}$ is the gravitational constant, $f \in \mathbb{R}$ is the total thrust, $m \in \mathbb{R}$ is the mass of the QUAV, and $\phi \in (-\pi/2, \pi/2)$, $\theta \in (-\pi/2, \pi/2)$, and $\psi \in (-\pi, \pi)$ denote the roll, pitch, and yaw angles, respectively.

Therefore, we present the control structure as shown in Figure 10. The position loop in the horizontal channel is controlled by the formation control law proposed in this paper, while the attitude loop is controlled by a PD controller. Assume that all quadrotors move within the X - Y plane, with their altitude and yaw angle remaining constant, i.e., the altitude channel input $u_{zd} = 0$ and the yaw angle $\psi_d = 0$, where $(\cdot)_d$ represents the desired value. The input of the horizontal channel $\mathbf{u}_{hd} \in \mathbb{R}^2$ is updated at each time step according to eqs. (6), (43), and (56). Then, ϕ_d and θ_d can be obtained by inverting eq. (65). Ultimately, the rotational speeds $\varpi_{d,k}$ of the four rotors are obtained, where $k = 1, 2, 3, 4$.

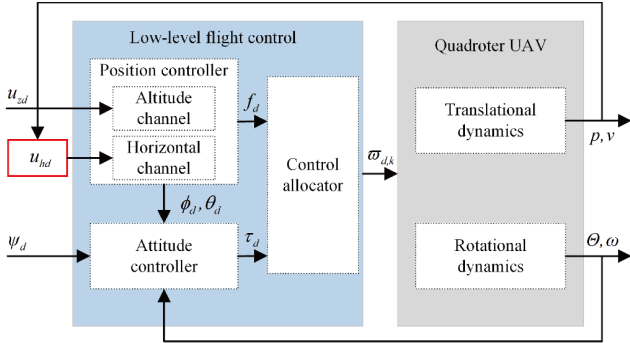


Figure 10 The control structure of the QUAVs.

4.1 Triangular formation maneuvering case

We first consider the triangular formation maneuvering case. The initial formation of the three quadrotors is an isosceles right triangle, where the initial position coordinates and velocity are selected as $\mathbf{p}_1(0) = [0, 0]^T$ m, $\mathbf{p}_2(0) = [4, 0]^T$ m, $\mathbf{p}_3(0) = [0, 4]^T$ m, $\mathbf{v}_1(0) = [0, 0]^T$ m/s, $\mathbf{v}_2(0) = [0, 0]^T$ m/s, $\mathbf{v}_3(0) = [0, 0]^T$ m/s. Then, let the desired formation configuration be the equilateral triangle, i.e., $\alpha_1^* = \alpha_2^* = \alpha_3^* = \pi/3$.

Applying the control law (6), the simulation results of the formation maneuvering are shown in Figure 11. The control gains are selected as $k_1 = 0.2, k_2 = 0.1, k_3 = 0.02$. The desired translational velocity is selected as $\mathbf{v}_c^* = [0.5, 0]^T$ m/s. And the desired distance d_{13}^* is given by

$$d_{13}^*(t) = \begin{cases} 10 \text{ m}, & 0 < t \leq 120 \text{ s}, \\ 5 \text{ m}, & 120 \text{ s} < t \leq 240 \text{ s}, \\ 10 \text{ m}, & t > 240 \text{ s}. \end{cases} \quad (68)$$

In Figure 11, the dotted line depicts the movement trajectories of the quadrotors. First, it shows the evolution from an initial formation to an equilateral triangle formation of the desired scale. Then, changing the desired distance d_{13}^* when the formation flying encounters narrow terrain allows the formation to maintain an equilateral triangle formation through the terrain. Finally, the scale of the formation is restored after passing the terrain. Figure 12 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (6).

4.2 Polygonal formation maneuvering case

In this case, we consider the desired formation configuration of six quadrotors. The initial configuration of the formation is given arbitrarily. The initial position coordinates and velocity are selected as $\mathbf{p}_1(0) = [0, 0]^T$ m, $\mathbf{p}_2(0) = [2, 2]^T$ m, $\mathbf{p}_3(0) = [-1, 4]^T$ m, $\mathbf{p}_4(0) = [5, 3]^T$ m, $\mathbf{p}_5(0) = [4, 0]^T$ m, $\mathbf{p}_6(0) =$

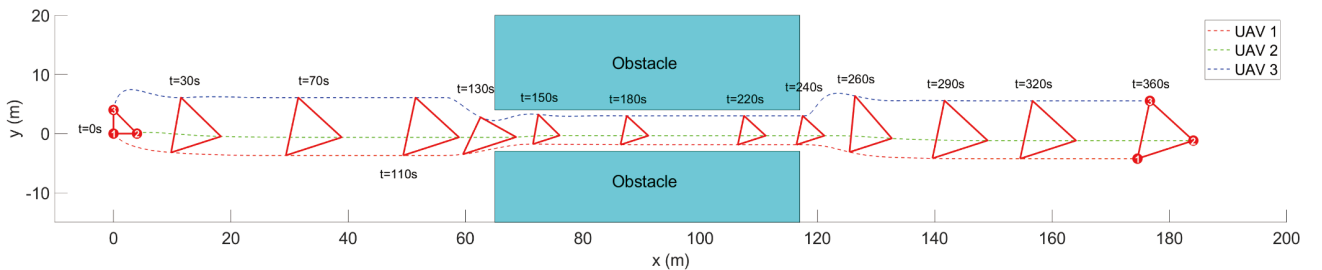


Figure 11 The triangle formation maneuvering trajectories when encountering narrow terrain.

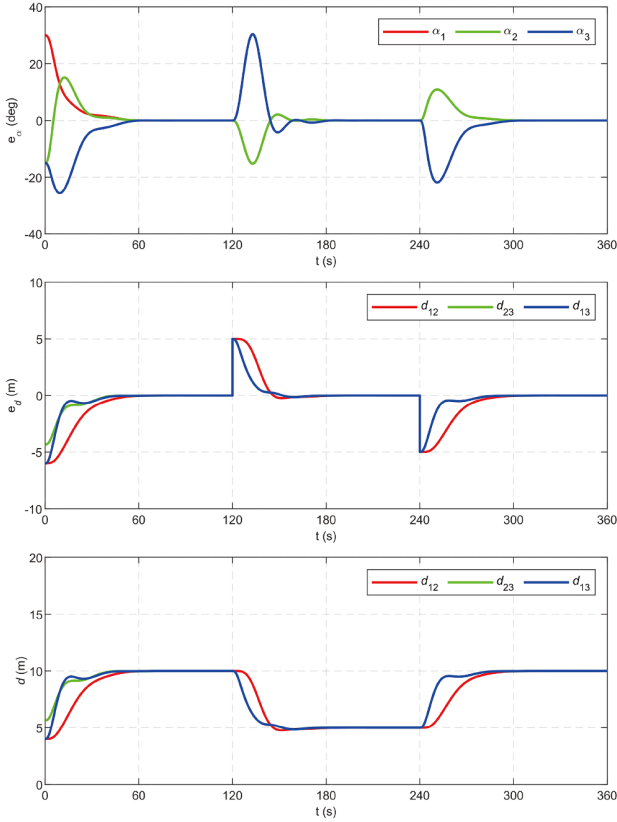


Figure 12 Angle and distance error trajectories and actual distance evolution trajectories for the triangular formation case.

$[4, 5]^T$ m, $\mathbf{v}_1(0) = [0, 0]^T$ m/s, $\mathbf{v}_2(0) = [0, 0]^T$ m/s, $\mathbf{v}_3(0) = [0, 0]^T$ m/s, $\mathbf{v}_4(0) = [0, 0]^T$ m/s, $\mathbf{v}_5(0) = [0, 0]^T$ m/s, $\mathbf{v}_6(0) = [0, 0]^T$ m/s. Then, let the desired formation configuration be the square, i.e., $\alpha_{213}^* = \pi/6$, $\alpha_{123}^* = 2\pi/3$, $\alpha_{132}^* = \pi/6$, $\alpha_{142}^* = \pi/6$, $\alpha_{243}^* = \pi/6$, $\alpha_{152}^* = \pi/3$, $\alpha_{254}^* = \pi/3$, $\alpha_{263}^* = \pi/3$, $\alpha_{264}^* = \pi/3$.

Applying the control law (43), the simulation results of the formation maneuvering are shown in Figure 13. The control gains are selected as $k_1 = 0.3$, $k_2 = 0.4$, $k_3 = 0.03$. The desired translational velocity is selected as $\mathbf{v}_c^* = [0.5, 0]^T$ m/s. The desired distance d_{13}^* is given by eq. (68).

First, Figure 13 shows the evolution from the initial formation to a polygonal formation of the desired size. Then,

when the formation encounters narrow terrain during flight, by changing the desired distance d_{13}^* , the formation is able to maintain its original geometric topology while passing through the terrain. Finally, the formation restores its size after passing through the terrain. Figure 14 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (43).

4.3 Square formation maneuvering control on a circle case

In this case, we consider the desired formation configuration of four quadrotors on a circle. The initial configuration of the formation is given arbitrarily. The initial position coordinates and velocity are selected as $\mathbf{p}_1(0) = [0, 0]^T$ m, $\mathbf{p}_2(0) = [4, 0]^T$ m, $\mathbf{p}_3(0) = [6, 2]^T$ m, $\mathbf{p}_4(0) = [0, 4]^T$ m, $\mathbf{v}_1(0) = [0, 0]^T$ m/s, $\mathbf{v}_2(0) = [0, 0]^T$ m/s, $\mathbf{v}_3(0) = [0, 0]^T$ m/s, $\mathbf{v}_4(0) = [0, 0]^T$ m/s. Then, let the desired formation configuration be the square, i.e., $\alpha_1^* = \alpha_2^* = \alpha_3^* = \alpha_4^* = \pi/2$.

Applying the control law (56), the simulation results of the formation maneuvering are shown in Figure 15. The control gains are selected as $k_1 = 0.5$, $k_2 = 0.5$, $k_3 = 0.1$. The desired translational velocity is selected as $\mathbf{v}_c^* = [0.5, 0]^T$ m/s. The desired distance d_{32}^* and d_{34}^* are given by

$$d_{32}^*(t) = d_{34}^*(t) = \begin{cases} 10 \text{ m}, & 0 < t \leq 120 \text{ s}, \\ 5 \text{ m}, & 120 \text{ s} < t \leq 240 \text{ s}, \\ 10 \text{ m}, & t > 240 \text{ s}. \end{cases} \quad (69)$$

First, Figure 15 shows the evolution from the initial formation to a square formation of the desired size. Then, when the formation encounters narrow terrain during flight, by changing the desired distances d_{32}^* and d_{34}^* , the formation is able to maintain its original square shape while passing through the terrain. Finally, the formation restores its size after passing through the terrain. Figure 16 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (56).

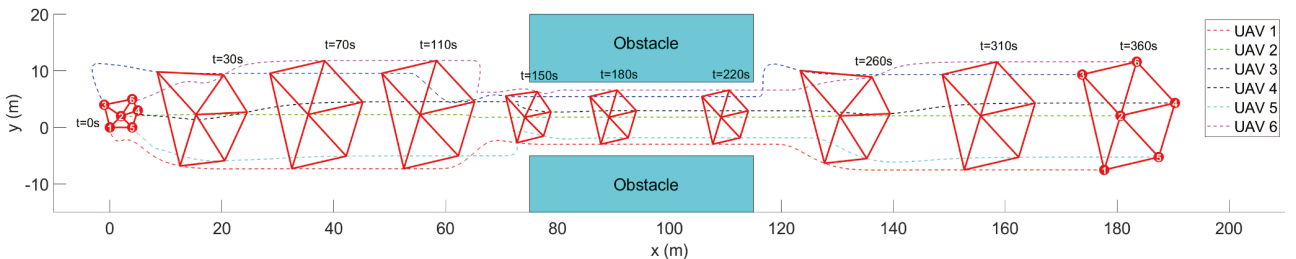


Figure 13 The polygonal formation maneuvering trajectories when encountering narrow terrain.

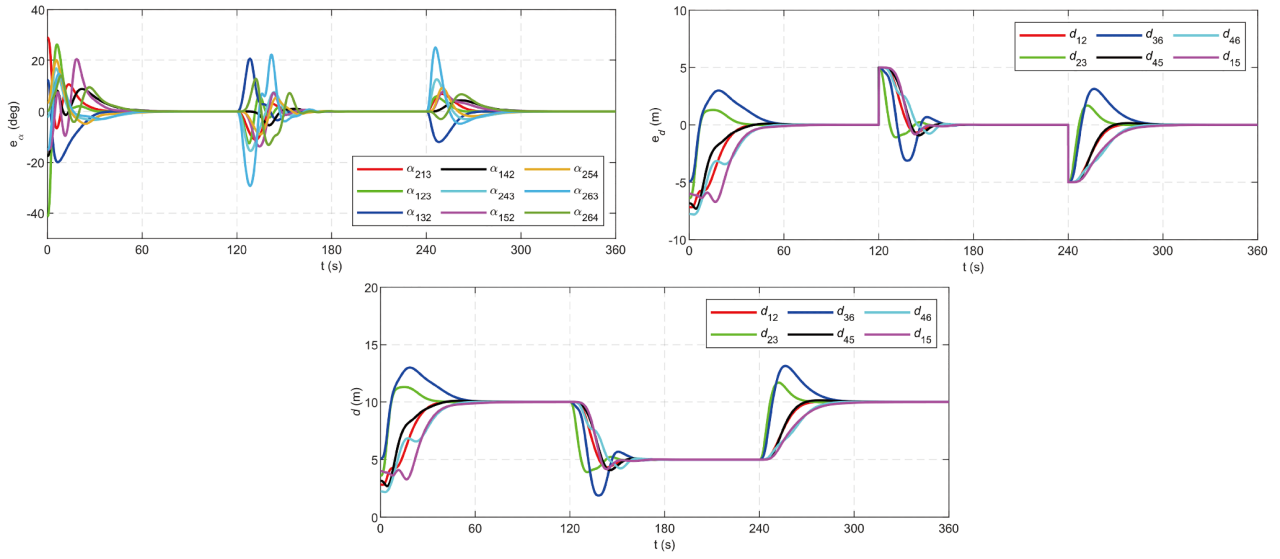


Figure 14 Angle and distance error trajectories and actual distance evolution trajectories for the polygonal formation case.

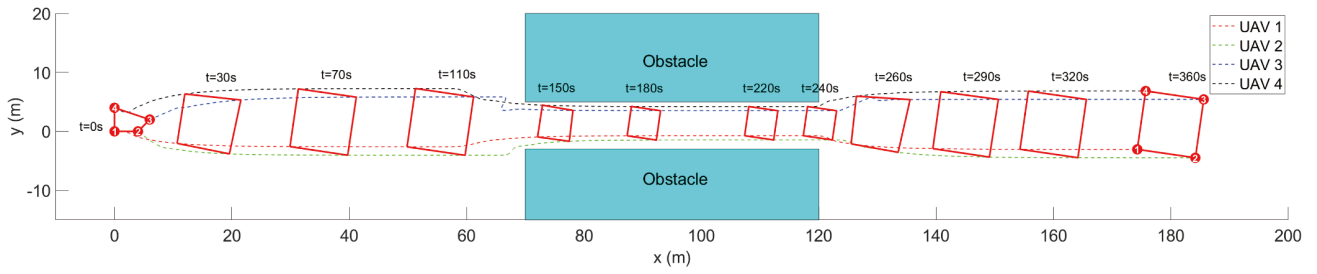


Figure 15 The square formation maneuvering trajectories when encountering narrow terrain.

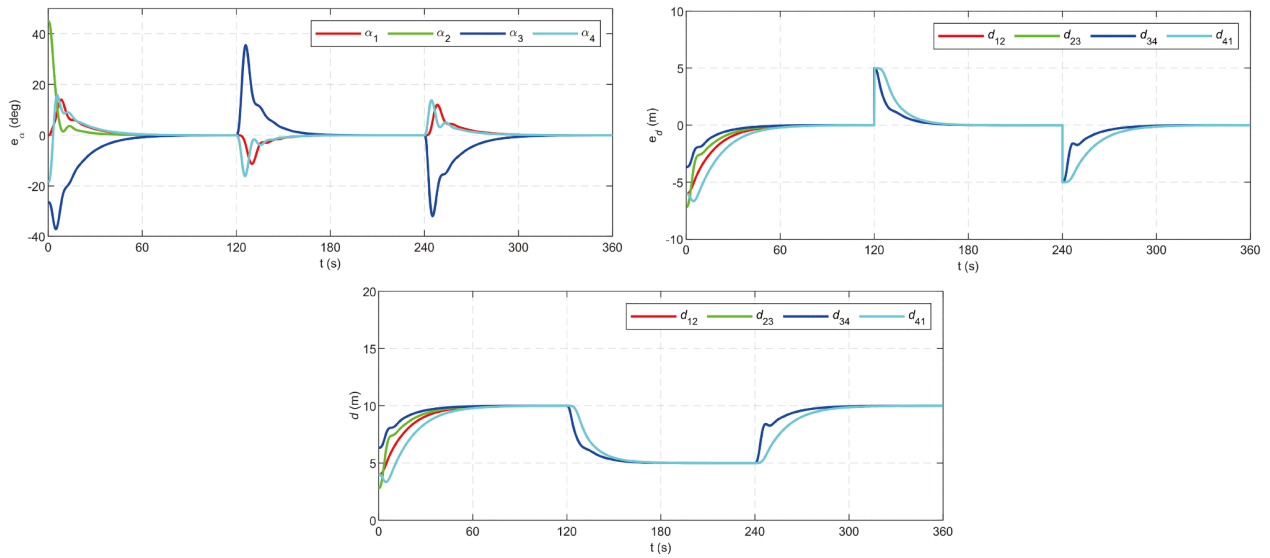


Figure 16 Angle and distance error trajectories and actual distance evolution trajectories for the square formation case.

It can be seen from Figures 11, 13, and 15 that the direction of the entire formation relative to the global coordinate system after deformation and expansion is uncertain, precisely because we are using local information. However, in all, the purpose of scaling the deformation and formation has been achieved. In addition, when the formation encounters obstacles that require scaling transformations, we only consider maintaining the size of the bounding box to ensure safe passage through narrow spaces.

5 Conclusion and discussion

This paper investigates formation control problems in multi-agent systems and designs control laws that regulate deformation and formation scaling using angle and distance information. Each agent measures the angle, with only one agent measuring the distance. This approach effectively addresses the ambiguity in controlling the scale of the formation when relying solely on angle information. The overall scaling of the formation is controlled by a single agent's distance measurement. When a rigid formation scales to avoid obstacles, it does not require extensive adjustments of distance constraints between different agents. Multiple experiments have been conducted to demonstrate the effectiveness of the proposed control algorithms. The results show that agent formations can navigate through narrow terrains with flexible sizes.

Currently, we can only prove the local convergence of the closed-loop nonlinear formation double integrator dynamics; global convergence remains unknown. Although the method is feasible for n agents, its performance and complexity may be affected as the number of agents increases significantly. Due to the diverse geometric shapes of formations with different numbers of agents, discussing all possible configurations in detail is impractical. For obstacle avoidance tasks, the primary goal is to avoid obstacles, and adopting a feasible formation scheme is sufficient without considering arbitrary formation shapes. Future work will address global convergence issues.

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Appendix A

For Section 3, see inequality (41), substituting a_3, a_2, a_1 , and a_0 into inequality (41) and organizing, we have

$$\begin{aligned} \Delta_2 = & \begin{vmatrix} a_2 & a_3 \\ a_0 & a_1 \end{vmatrix} \\ = & (k_2^3 f_{23} + k_2^2 k_3 p) g_1^2 + 2k_2^2 k_3 p g_1 g_2 \\ & + k_2^2 k_3 g_1 [f_{13}(p - q) + 2f_{23}p + f_{13}p] + k_2^2 k_3 p^2 g_1 \\ & + (k_2^3 f_{13} + k_2^2 k_3 p) g_2^2 + k_2^2 k_3 f_{13}^2 (p - q) \\ & + k_2^2 k_3 (2f_{13}p + f_{23}p) g_2 + k_2^2 k_3 p^2 g_2 + k_2^2 k_3 f_{13} f_{21} p \\ & + 2k_2^3 (f_{13} + f_{23}) g_1 g_2 + k_2^3 g_1^2 g_2 + k_2^3 g_1 g_2^2 \\ & + k_2^3 g_1 (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23} + f_{13} f_{23} + f_{23}^2) \\ & + k_2^3 g_2 (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23} + f_{13} f_{23} + f_{13}^2) \\ & + k_2^3 (f_{13} + f_{23}) (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23}) \\ & + k_2^2 k_3 f_{13} f_{23} p - k_2^2 k_3 f_{13} f_{23} q + k_2^2 k_3 f_{13} p (p - q) \\ & + k_2^2 k_3 f_{12} f_{23} q, \end{aligned} \quad (\text{a1}) \end{aligned}$$

where for each term with a negative sign in eq. (a1), a positive term can be found and the result of adding it is positive by using $p > 0$, $p - q > 0$, and $g_1 g_2 - f_{12} f_{21} > 0$, except for the last term whose sign is uncertain.

Now, let us analyze the positive or negative of the last term. Considering $k_2^2 k_3 f_{12} f_{23} q$, there are the following cases that determine the positivity of eq. (a1).

Case 1 If

$$d_{13} < d_{12} < d_{23}, \quad (\text{a2})$$

or

$$d_{12} < d_{13} < d_{23}, \quad (\text{a3})$$

or

$$d_{12} < d_{23} < d_{13}, \quad (\text{a4})$$

we have $\alpha_3 < \alpha_1$ and $\cos \alpha_3 > \cos \alpha_1$, as a result, $q > 0$, then

$\Delta_2 > 0$.

Case 2 If

$$d_{13} < d_{23} < d_{12}, \quad (\text{a5})$$

or

$$d_{23} < d_{13} < d_{12}, \quad (\text{a6})$$

we have $\alpha_3 > \alpha_1$ and $\cos \alpha_3 < \cos \alpha_1$, then $q < 0$. But by using the law of sines (54), one has

$$\begin{aligned} f_{13} - f_{12} = & \frac{\sin \alpha_3}{d_{13}} - \frac{\sin \alpha_2}{d_{12}} = 2R \frac{d_{12}^2 - d_{13}^2}{d_{12} d_{13}} > 0, \\ & -f_{13} f_{23} q > -f_{12} f_{23} q, \\ & -f_{13} f_{23} q + f_{12} f_{23} q > 0, \\ & -k_2^2 k_3 f_{13} f_{23} q + k_2^2 k_3 f_{12} f_{23} q > 0, \end{aligned} \quad (\text{a7})$$

so that $\Delta_2 > 0$.

Case 3 If

$$d_{23} < d_{12} < d_{13}, \quad (\text{a8})$$

we have

$$f_{32} - f_{12} = \frac{\sin \alpha_2}{d_{23}} - \frac{\sin \alpha_2}{d_{12}} > 0. \quad (\text{a9})$$

There are two cases: when α_1, α_2 and α_3 are all acute angles, one has

$$p + q = 1 + \cos \alpha_3 + \cos \alpha_3 - \cos \alpha_1 > 0. \quad (\text{a10})$$

And when α_2 is an obtuse angle, α_1 and α_3 are acute angles, there is also

$$p + q > 0, \quad (\text{a11})$$

where we obtained

$$\begin{aligned} f_{12} f_{23} p + f_{12} f_{23} q & > 0, \\ f_{32} f_{23} p + f_{12} f_{23} q & > 0. \end{aligned} \quad (\text{a12})$$

By using $g_2 = f_{32} + f_{12}$, one has

$$k_2^2 k_3 f_{32} f_{23} p + k_2^2 k_3 f_{12} f_{23} q > 0, \quad (\text{a13})$$

where we have $\Delta_2 > 0$.

In conclusion, it has been proven that $\Delta_2 > 0$ is valid.